AN OPEN MAP WITH IRREDUCIBLE TARGET AND IRREDUCIBLE FIBRES OVER A DENSE SET OF POINTS HAS IRREDUCIBLE SOURCE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of irreducibility through the impossibility of writing the space as a union of two proper closed subsets, and the openness of the map. Set up in the author's section on irreducible components, immediately before the counted statement. Fixes the three hypotheses (a), (b), (c) and the conclusion.

Lemma 0.1. *Let $f : X \to Y$ be a continuous map of topological spaces. Assume that (a) Y is irreducible, (b) f is open, and (c) there exists a dense collection of points $y \in Y$ such that $f^{-1}(y)$ is irreducible. Then X is irreducible.*

Proof. Suppose $X = Z_1 \cup Z_2$ with Z_i closed. Consider the open sets $U_1 = Z_1 \setminus Z_2 = X \setminus Z_2$ and $U_2 = Z_2 \setminus Z_1 = X \setminus Z_1$. To get a contradiction assume that U_1 and U_2 are both nonempty. By (b) we see that $f(U_i)$ is open. By (a) we have Y irreducible and hence $f(U_1) \cap f(U_2) \neq \emptyset$. By (c) there is a point y which corresponds to a point of this intersection such that the fibre $X_y = f^{-1}(y)$ is irreducible. Then $X_y \cap U_1$ and $X_y \cap U_2$ are nonempty disjoint open subsets of X_y which is a contradiction. $\square$

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